Programming and Artificial Intelligence — Part One

Chapter 3 — Web Applications

Questions & Answers — Exam-Style Practice

Lessons covered

- 3-1 — The Overall Structure of Web Applications
- 3-2 — Web Application Communication Methods
- 3-3 — Fundamentals of Frontend Technology

This practice document follows the Chapter 3 exercise style in the provided textbook: fill in the blanks, true/false, matching/classification, multiple choice, 6-mark questions, Think as an Engineer — Investigate & Decide, New Context, and Reflect & Challenge.

Lesson 3-1 — The Overall Structure of Web Applications

Key idea: a web application uses three cooperating tiers — frontend, backend, and database.

1. Read the following passage and answer each question.

A web application is broadly composed of three tiers. The screen component that users see directly is called the (a). The part responsible for server-side processing that users do not see is called the (b). A system that organizes, stores, and manages data is called the (c).

1. Fill in the blanks (a)–(c).

Answer: (a) Frontend (b) Backend (c) Database.

2. Mark each statement (A–D) with ■ if true or × if false.

A. The frontend is the screen component that users see and interact with directly.

Answer: ■ True

B. The backend performs server-side processing that users do not see.

Answer: ■ True

C. The database is mainly responsible for choosing the colors and layout of a web page.

Answer: × False — the frontend/CSS handles appearance; the database stores and manages data.

D. The three tiers cooperate to make a web application work.

Answer: ■ True

3. For each of a–f, indicate which tier it belongs to: A Frontend, B Backend, C Database.

a	Product listing page → A Frontend
b	Stock checking processing → B Backend
c	Storage of customer information → C Database
d	Cart screen → A Frontend
e	Payment processing → B Backend
f	Storage of order history → C Database

4. Choose the one that most appropriately describes the frontend.

A. A system that stores customer information
B. The screen component users see and interact with directly
C. A server-side processing system
D. A communication protocol

Answer: B.

5. Complete the application flow.

A user enters a product search keyword. The (a) receives the keyword and sends it to the (b). The backend searches the (c). The database returns matching data. The backend organizes the result and sends it to the frontend, which (d) the result.

Answer: (a) frontend; (b) backend; (c) database; (d) displays/shows.

6. Short-answer questions

Why is the frontend important?

Answer: It is the part users see and interact with directly, and it collects user input.

What is the role of the backend?

Answer: It performs server-side processing, processes data, and makes decisions.

What is the role of the database?

Answer: It organizes, stores, manages, retrieves, and records data.

Why must the three tiers cooperate?

Answer: Each tier has a clear responsibility, and together they complete the user's request and result.

7. 6 Marks

Explain how the three tiers of a web application cooperate when a user posts a photo on a social media app. [6] Refer to: the role of the frontend, backend, and database.

Answer: The frontend provides the photo-upload screen and collects the user's action. The backend receives the request and performs server-side processing. The database stores and manages information related to the post and user. The three tiers cooperate so the photo can be submitted, processed, stored, and later displayed.

8. Think as an Engineer — Investigate & Decide

A clinic wants a web app where patients book appointments, check which time slots are free, and see their past visits.

1 · Assign the work: which tier handles the booking form, checking free time slots, and storing past visits?

Answer: Booking form → frontend. Checking available slots → backend using data from the database. Storing past visits → database.

2 · The booking page opens normally, but no free slots appear. Which tier is clearly working, and which one has probably failed?

Answer: The frontend is clearly working because the page opens. The backend or database may have failed because the slot information is not being processed or retrieved.

3 · Decide: should the clinic separate the app into three tiers? Give two reasons.

Answer: Yes. Separation gives each tier a clear responsibility and makes development, maintenance, and troubleshooting easier.

9. New Context

A weather app lets users type a city and see the forecast. Using the three-tier model, name the tier that takes the city name, looks up the forecast data, and stores past forecasts. Identify one job the frontend cannot do on its own.

Answer: Taking the city name → frontend. Looking up/processing forecast data → backend. Storing past forecasts → database. The frontend cannot perform the server-side processing or manage the stored database data on its own.

10. Reflect & Challenge

Reflect: Why would putting frontend, backend, and database responsibilities into one undivided part make a large web application harder to manage?

Answer: Different responsibilities would be mixed together, making development, maintenance, and troubleshooting harder.

Challenge: Name one web application you use and describe one task done by each tier.

Answer: Example: an online shopping site — frontend shows products and collects searches; backend processes searches/orders; database stores product, customer, and order information.

Lesson 3-2 — Web Application Communication Methods

Key idea: client requests, server responds — over HTTP/HTTPS, with methods, status codes, APIs, and often JSON.

1. Read the following passage and answer each question.

A web application operates by communication between a (a), which requests data or processing from the server, and a (b), which receives the request, processes it, and returns a response. This communication uses a protocol called (c). The protocol that adds SSL/TLS encryption to (c) is called (d).

1. Fill in the blanks (a)–(d).

Answer: (a) Client; (b) Server; (c) HTTP; (d) HTTPS.

2. Mark each statement (A–D) with ■ if true or × if false.

A. A browser can act as a client.

Answer: ■ True

B. A server receives requests, processes them, and returns responses.

Answer: ■ True

C. HTTPS removes encryption from HTTP.

Answer: × False — HTTPS adds SSL/TLS encryption to HTTP.

D. GET retrieves data, while POST sends data to a server.

Answer: ■ True

3. For each operation (a–d), indicate which HTTP method is used: A GET, B POST.

a	Displaying a web page → A GET
b	Entering an ID and password into a login form and submitting it → B POST
c	Displaying search results → A GET
d	Submitting the contents of a contact form → B POST

4. From the following options, choose the one that most appropriately describes HTTPS.

A. A method for retrieving data
B. A system for exchanging data between software
C. A communication protocol that adds SSL/TLS encryption to HTTP
D. A status code indicating a server problem

Answer: C.

5. Status codes

What does 200 indicate?

Answer: The request was processed successfully.

What does 404 indicate?

Answer: The requested page was not found / the URL does not exist.

What does 500 indicate?

Answer: A server-side error occurred.

6. APIs and JSON

What is an API?

Answer: A system for exchanging data between different pieces of software.

What does JSON stand for?

Answer: JavaScript Object Notation.

Why is JSON useful in web applications?

Answer: It is a data format that is easy for humans to read and computers to process.

7. Multiple choice

Which HTTP method retrieves a forecast?

A. GET   B. POST   C. 404   D. HTTPS

Answer: A

Which method is appropriate for submitting a contact form?

A. GET   B. POST   C. 404   D. 500

Answer: B

Which status code means the requested page was not found?

A. 200   B. 404   C. 500   D. 100

Answer: B

Which status code indicates a server-side error?

A. 200   B. 201   C. 404   D. 500

Answer: D

Which statement best describes an API?

A. A screen design method
B. A system for exchanging data between different pieces of software
C. A database table
D. A status code

Answer: B

8. 6 Marks

Explain how a browser and a server communicate when a user opens a web page. [6] Refer to client, server, protocol, request, and response.

Answer: The browser acts as the client and sends a request to the server. The communication uses HTTP or HTTPS. The server receives and processes the request and returns a response. The browser receives the response and displays the requested page.

9. Think as an Engineer — Investigate & Decide

A student is building a login page and wants passwords protected while being sent to the server.

1 · Should the page use HTTP or HTTPS? Give a reason.

Answer: HTTPS, because it adds SSL/TLS encryption and helps protect communication content.

2 · Which HTTP method is appropriate for sending the login data?

Answer: POST.

3 · The page shows 'Not Found.' Which status code would help explain the problem?

Answer: 404.

10. New Context

A weather service offers an API that returns today's forecast when a client sends a request. Name the HTTP method used to retrieve the forecast, why HTTPS should be used, and one advantage of JSON.

Answer: GET retrieves the forecast. HTTPS adds SSL/TLS encryption. JSON is easy for humans to read and computers to process.

11. Reflect & Challenge

Reflect: When a page fails to load, which status code would help you understand why, and what would it tell you?

Answer: 404 indicates a missing page/URL; 500 indicates a server-side error. The correct code depends on the problem.

Challenge: Name one app you use and describe one request it sends and the response it expects.

Answer: Example: a weather app sends a request for a city's forecast and expects forecast data in the response.

Lesson 3-3 — Fundamentals of Frontend Technology

Key idea: HTML structures, CSS styles, JavaScript behaves; semantic HTML, responsive design, and frameworks strengthen the frontend.

1. Read the following passage and answer each question.

Web pages are mainly composed of three technologies: (a), which defines the structure of the page; (b), which specifies the appearance; and (c), which adds behavior and interactivity. The concept of adjusting the layout according to different screen sizes such as PCs, smartphones, and tablets is called (d).

1. Fill in the blanks (a)–(d).

Answer: (a) HTML; (b) CSS; (c) JavaScript; (d) Responsive design.

2. Mark each statement (A–D) with ■ if true or × if false.

A. HTML defines the structure of a web page.

Answer: ■ True

B. CSS adds behavior to a web page.

Answer: × False — JavaScript adds behavior and interactivity.

C. Responsive design adjusts the layout according to different screen sizes.

Answer: ■ True

D. A framework can make development more efficient by providing commonly used functions and components.

Answer: ■ True

3. Match each role (a–c) with A HTML, B CSS, C JavaScript.

a	Defines structure → A HTML
b	Specifies appearance → B CSS
c	Adds behavior and interactivity → C JavaScript

4. For each task (a–f), indicate A HTML, B CSS, or C JavaScript.

a	Headings and paragraphs → A HTML
b	Changing text color and size → B CSS
c	Updating part of the screen when a button is pressed → C JavaScript
d	Placing images and links → A HTML
e	Setting the background color → B CSS
f	Validating form contents → C JavaScript

5. Semantic HTML

What is semantic HTML?

Answer: Using element names/tags that indicate the meaning of each part of a web page, such as , , , and .

Name two benefits.

Answer: Improved accessibility for screen readers and improved SEO because search engines can better understand page content.

6. Choose the one that is NOT an appropriate benefit of semantic HTML.

A. Screen readers can better understand the structure.
B. Search engines can more easily understand content.
C. The display speed necessarily becomes faster.
D. Information becomes easier to convey to users with visual impairments.

Answer: C.

7. Responsive design and mobile-first

What is responsive design?

Answer: A design concept in which the layout is automatically adjusted according to different screen sizes.

Why is responsive design needed?

Answer: Websites are accessed from different devices such as PCs, smartphones, and tablets.

What is mobile-first?

Answer: Prioritizing small screens such as smartphones from the design stage.

Choose the reason responsive design is needed: A. Improve display speed B. Support various devices C. Raise search rankings D. Add behavior.

Answer: B.

8. Frameworks

What is a framework?

Answer: A structure that provides commonly used functions and components in advance to make web application development more efficient.

Choose the best description: A. Language for page structure B. Structure for efficient web application development C. Encryption technology D. Screen-size design concept.

Answer: B.

9. Major frontend frameworks

React	Developed by Meta; can efficiently update only part of the screen and is suited to large-scale web applications.
Vue	Simple structure; allows applications to be built incrementally from small features.
Next.js	Adds mechanisms to speed up page display in addition to React's features.

10. 6 Marks

Explain how HTML, CSS, and JavaScript each contribute to a library page that lists books, is styled, and filters results as the user types. [6]

Answer: HTML defines the structure and content. CSS specifies appearance such as colors, layout, fonts, and sizes. JavaScript adds behavior such as filtering results while the user types and updating part of the screen.

11. Think as an Engineer — Investigate & Decide

A student club wants a page that lists events, uses the club's colors, and has a button showing how many people have signed up.

1 · Assign each feature to HTML, CSS, or JavaScript.

Answer: Event list → HTML. Colors and layout → CSS. Live sign-up counter → JavaScript.

2 · Name one responsive-design change and one semantic-HTML choice that would help a phone user and a screen reader.

Answer: Use a responsive layout for the phone. Use meaningful elements such as , , or for the screen reader.

3 · Should the club separate HTML, CSS, and JavaScript? Give two reasons.

Answer: Yes. Each technology has a clear role, and separation makes the page easier to understand, maintain, and develop.

12. New Context

A news site must look good on phones and be found easily by search engines. Give one responsive-design choice and one semantic-HTML choice, and state what each improves.

Answer: Use a responsive layout that re-flows content for small screens, improving usability. Use semantic elements such as , , and , improving accessibility and helping search engines understand the content.

13. Reflect & Challenge

Reflect: Of structure, style, and behavior, which do you notice most when a web page is badly made, and why?

Answer: Any can be noticeable: poor structure can confuse users, poor style can reduce readability, and poor behavior can make controls fail to respond.

Challenge: Name one website you use and describe one thing HTML, CSS, and JavaScript each do on it.

Answer: Example: HTML provides headings and links; CSS controls layout and appearance; JavaScript responds to clicks, typing, or other interactions.

Chapter 3 — Mixed Revision Test

Try these questions without looking at the answers.

Part A — Fill in the blanks

1. Frontend + backend + ________ make the three-tier architecture.

   Answer: database

2. The side that requests data or processing is the ________.

   Answer: client

3. The communication protocol is ________.

   Answer: HTTP

4. HTTP with SSL/TLS encryption is ________.

   Answer: HTTPS

5. The method used to retrieve data is ________.

   Answer: GET

6. The method used to send data is ________.

   Answer: POST

7. Status code ________ means success.

   Answer: 200

8. Status code ________ means not found.

   Answer: 404

9. Status code ________ indicates a server-side error.

   Answer: 500

10. ________ defines structure.

    Answer: HTML

11. ________ specifies appearance.

    Answer: CSS

12. ________ adds behavior.

    Answer: JavaScript

13. Layout adjustment for different screen sizes is ________.

    Answer: responsive design

Part B — True / False

1. The database is the part users directly see.

   Answer: × False

2. A browser can act as a client.

   Answer: ■ True

3. HTTPS adds SSL/TLS encryption to HTTP.

   Answer: ■ True

4. GET is the appropriate method for submitting a contact form.

Answer: × False — POST is appropriate.

5. 404 means the requested page was not found.

Answer: ■ True

6. JSON is easy for humans to read and computers to process.

Answer: ■ True

7. CSS is mainly responsible for behavior.

Answer: × False — JavaScript handles behavior.

8. Responsive design supports different screen sizes.

Answer: ■ True

9. Semantic HTML can improve accessibility and SEO.

Answer: ■ True

10. Frameworks require developers to write every common function from scratch.

Answer: × False

Part C — Matching

Match each item with A Frontend, B Backend, or C Database.

a	Product page → A Frontend
b	Order processing → B Backend
c	Stored customer information → C Database
d	Cart screen → A Frontend
e	Stock checking → B Backend
f	Order history → C Database

Match each item with A HTML, B CSS, or C JavaScript.

a	Headings and paragraphs → A HTML
b	Font size and colors → B CSS
c	Button changes screen → C JavaScript
d	Images and links → A HTML
e	Page layout → B CSS
f	Form validation → C JavaScript

Part D — Multiple Choice

1. Which part stores and manages data?

A. Frontend B. Backend C. Database D. Browser

Answer: C

2. Which HTTP method retrieves data?

A. GET B. POST C. 404 D. HTTPS

Answer: A

3. Which status code indicates success?

A. 200 B. 404 C. 500 D. 301

Answer: A

4. Which status code indicates a missing page?

A. 200   B. 404   C. 500   D. 100

Answer: B

5. Which technology defines structure?

A. CSS   B. HTML   C. JavaScript   D. JSON

Answer: B

6. Which technology controls appearance?

A. HTML   B. CSS   C. JavaScript   D. HTTP

Answer: B

7. Which technology adds behavior?

A. HTML   B. CSS   C. JavaScript   D. Database

Answer: C

8. Which is a benefit of semantic HTML?

A. Necessarily faster display   B. Better accessibility and SEO   C. Replaces JavaScript   D. Encrypts HTTP

Answer: B

9. Which concept adapts layout to screen size?

A. Responsive design   B. API   C. POST   D. Database

Answer: A

10. Which framework was developed by Meta?

A. Vue   B. React   C. Next.js   D. HTML

Answer: B

Part E — 6 Marks

1. Explain how a product search works through the three-tier architecture. [6]

Answer: Frontend collects the keyword; backend receives and processes the request; database is searched; matching data returns to the backend; backend organizes the result; frontend displays it.

2. Explain secure client-server communication when a user opens a web page. [6]

Answer: The browser is the client and sends a request. HTTPS can carry the request securely because it adds SSL/TLS encryption to HTTP. The server processes the request and returns a response, which the browser displays.

3. Explain HTML, CSS, and JavaScript on an interactive web page. [6]

Answer: HTML defines structure/content, CSS controls appearance, and JavaScript adds behavior and interactivity.

Part F — Think as an Engineer

A school wants a web application where students view events, submit registration forms, and receive confirmation.

1. Which tier displays the event list and registration form?

Answer: Frontend.

2. Which tier processes the registration request?

Answer: Backend.

3. Which tier stores registration information?

Answer: Database.

4. Which HTTP method is appropriate for submitting the form?

   Answer: POST.

5. Why use HTTPS?

   Answer: It adds SSL/TLS encryption and helps protect communication content.

6. Which status code is appropriate if the requested page does not exist?

   Answer: 404.

Part G — New Context

A small online library wants a page that works well on smartphones, uses meaningful HTML elements, has an interactive search box, and receives book data from a server. Identify: (1) screen-size concept, (2) meaningful-HTML concept, (3) interactive technology, (4) common data format.

Answer: (1) Responsive design. (2) Semantic HTML. (3) JavaScript. (4) JSON.

Part H — Reflect & Challenge

A page reports either that a resource cannot be found or that a server failed. Which two status codes would you investigate?

Answer: 404 for not found; 500 for a server-side error.

Challenge: Describe one complete journey from a user's action to the final result using at least five terms from: frontend, backend, database, client, server, HTTP/HTTPS, GET/POST, status code, API, JSON.

Answer: Example: The client uses the frontend to enter a search. The frontend sends a GET request over HTTPS to the server. The backend processes it and uses the database. The server returns a response, possibly through an API using JSON, and the frontend displays the result.

Teacher Answer Guide — Chapter 3 Memory Map

3-1	Frontend = shows/collects; Backend = decides/processes; Database = stores/retrieves.
3-2	Client requests; Server responds; HTTP communicates; HTTPS adds SSL/TLS encryption.
Methods	GET = retrieve; POST = send.
Status	200 = success; 404 = not found; 500 = server-side error.
Data	API = software-to-software data exchange; JSON = readable data format.
3-3	HTML = structure; CSS = appearance; JavaScript = behavior/interactivity.
Modern frontend	Semantic HTML → accessibility + SEO; Responsive design → adapts to screen size; Mobile-first → prioritizes small screens.
Frameworks	React = Meta; Vue = simple/incremental; Next.js = React features + mechanisms to speed page display.

Source basis: Chapter 3 of the provided Egyptian Baccalaureate textbook, lessons 3-1, 3-2, and 3-3, including the chapter's exercise and exam-style question formats.